Venkata Lakshmi Parimala Pasupuleti

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | Portfolio | Virginia, USA

PROFESSIONAL SUMMARY

Forward deployed engineer with 8+ years embedding with operational teams, scoping complex problems, and deploying production AI systems from initial design through go-live. At Entrata, I was the de facto technical owner for affordable housing compliance deployments -- scoping requirements with cross-functional stakeholders, building the modules, integrating with downstream systems, and owning production outcomes for 100K+ users. Claims-style workflows are my home domain: verify information, apply rules, make eligibility decisions, escalate exceptions to humans. I build multimodal agentic AI systems that automate exactly this workflow: LeaseI is an 8-agent LangGraph pipeline for compliance investigation (93.6% precision); AgentProbe ships a production eval harness used to validate it. Python, FastAPI, LangChain, LangGraph, prompt engineering, and enterprise integrations -- production-grade. M.S. Data Analytics Engineering, GMU (GPA 3.87). AgentProbe: github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

Customer-Embedded AI Deployment | Claims & Compliance Workflow Automation | Multimodal Agentic AI Systems | LangChain & LangGraph | Enterprise API Integration | HITL Design & AI Guardrails | Prompt Engineering & LLM Orchestration | Production Code Ownership

WORK EXPERIENCE

Blue Horse Digital

Sep 2026 -- Present

AI/ML Engineer (Volunteer) -- Virginia, USA

- Building ML models and agentic AI pipelines; integrating data systems with back-end APIs; iterating based on real application behavior and team feedback.

Magizhs Tech Solution

May 2025 -- Aug 2025

AI/ML Engineer Intern (Forward Deployed)

- **Scoped customer needs, designed solution, built and deployed end-to-end:** translated fuzzy marketing and real estate survey requests into a LangGraph agentic pipeline (parse -> plan -> retrieve -> validate -> conditional routing) -- delivering structured, compliant flight plans automatically.
- Integrated a RAG layer (FAA Part 107 + operational corpus) for grounded compliance decisions; engineered guardrails blocking non-compliant outputs before they reached users; primary technical contact across all delivery phases.

George Mason University

Aug 2024 -- May 2026

Graduate Teaching Assistant + Research Associate

- Built multi-source prediction pipelines (Python, R, R2 = 0.977); applied Isolation Forest for automated data integrity checks; designed Tableau/Quarto dashboards for faculty leadership (+40% comprehension).
- Fine-tuned a custom classification model (HuggingFace Transformers + PyTorch); deployed inference API on Quart; presented evaluation results to faculty stakeholders.

Entrata (PropTech / Real Estate SaaS)

Jan 2017 -- Aug 2024

Senior Software Engineer -- Lehi, UT (Remote) [promoted from Associate, 2018]

- **Acted as primary technical contact for affordable housing compliance deployments** -- embedded with operational, product, and legal stakeholders to scope requirements, build solutions, integrate across 5+ dependent systems, and own production outcomes for 100K+ monthly active users.
- **Owned claims-style eligibility and verification workflows end-to-end for 7 years:** annual income-limit imports (few thousand federal rows -> 180,000 domain records), rent and HAP calculations, eligibility determinations -- every decision documented, auditable, and compliant with HUD/LIHTC regulations.
- Built and owned the RabbitMQ-based async processing pipeline for gross-rent updates -- designed retry logic and idempotency guarantees; diagnosed and resolved production reliability issues; zero compliance violations across all annual cycles.
- Designed and integrated RESTful APIs and microservices across internal platforms; led AWS S3 migration with security controls (Libsodium, 100% PHI compliance, zero incidents); eliminated 80% of manual operational workload via Python/Bash automation pipelines.
- Mentored 5+ engineers; drove technical standards via 300+ code reviews; built internal debugging tools cutting team debugging time by 40%.

PROJECTS

LeaseI: Agentic Claims-Style Compliance Investigation Platform

LangGraph 8-agent -- HITL -- FastAPI -- RAG -- Multimodal

Built a production agentic AI platform that automates claims-style workflows: verify information, apply rules, determine eligibility, escalate exceptions. Three surfaces: **Investigation** (8-agent LangGraph pipeline with intent classification, multi-source retrieval, reasoning, and HITL escalation when confidence too low); **Document Intelligence** (multimodal OCR + TF-IDF RAG, citation grounding, human-review gate on regulated decisions); **Tenant Self-Service** (guardrail: AI never resolves eligibility -- always routes to staff). Results: 93.6% precision / 87.9% F1 on 1,240 lease PDFs; 100% citation correctness on Phase 1 eval set.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe: AI Agent Eval Harness & Orchestration Toolkit LangGraph -- LangChain -- FastAPI -- ChromaDB -- MCP

Production eval harness for multimodal agentic AI: 15-endpoint FastAPI backend, 5-metric LLM-as-judge engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI/CD regression gate blocking failing agent runs automatically, multi-provider LLM Router (OpenAI/Anthropic/Groq/Ollama), prompt engineering pipeline, observability dashboard. Used to validate LeaseI across 1,240+ compliance test cases.

github.com/venkatalakshmiparimala/AgentProbe

AI on FHIR: NLP-Driven Clinical Records Query System Flask -- LLMs -- BERT -- React -- Docker -- FHIR

Scoped with clinical staff, designed, and deployed an NLP platform translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings; HIPAA-compliant RBAC; reduced manual retrieval time ~60%; drove staff adoption across clinical workflows. Served as primary technical contact through go-live.

github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering -- George Mason University May 2026

GPA 3.87 -- Virginia, USA

B.Tech Computer Science & Engineering -- JNTU Anantapur 2017

SKILLS

AI & Agents: LangChain, LangGraph, Multimodal Agentic AI, Intent Classification, Tool Integration, Orchestration, Multi-Step Reasoning, HITL, AI Guardrails, Safety Controls, Content Filtering, Responsible AI, Prompt Engineering, RAG, Embeddings, LLM Routing, OpenAI API, Anthropic API, FastAPI, Observability, Hallucination Detection, MCP

Languages & Frameworks: Python, JavaScript, TypeScript, Node.js, Java, SQL, React.js, Django, Spring Boot, Flask, Streamlit, PHP

Cloud & Infra: AWS (S3, EC2, Lambda, Glue, Bedrock), Docker, Kubernetes, GitHub Actions CI/CD, PostgreSQL, RabbitMQ, Redis, Microservices, Distributed Systems

ML & Data: PyTorch, HuggingFace Transformers, scikit-learn, XGBoost, Isolation Forest, Pandas, NumPy, Tableau